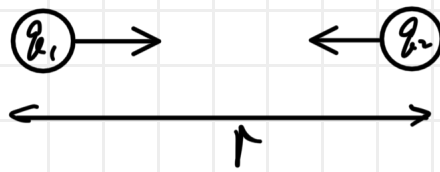


PHYS 212 by Snowy222827

Lecture 1

1. Coulomb's Law:

库仑定律



$$F \propto \frac{q_1 q_2}{r^2}$$

* } Attract each other
 } Repel each other 互斥

Our notation: $\vec{F}_{1,2}$ is the force by 1 on 2 (by-on)
 $\hat{r}_{12}$ is the unit vector that points from 1 to 2

$$\vec{F}_{1,2} = \frac{k q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

ex. Two iron paperclips are separated by 3 meters. Then you remove 1 electron from each atom on the first paperclip and place it on the second one.

$$\vec{F} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

$$k = 9 \times 10^9 \text{ N m}^2 / \text{C}^2$$

$$\text{electron charge} = 1.6 \times 10^{-19} \text{ Coulombs}$$

$$N_A = 6.02 \times 10^{23}$$

$$\text{Iron molar mass} \sim 56 \text{ grams/mol}$$

$$\text{mass of paper clip} \sim 1 \text{ grams}$$

r^- , k^-

mol: $\frac{1}{56}$ mol

What will the direction of the force be?

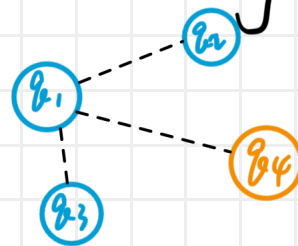
A) Attractive

B) Repulsive

$$F = k \frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{(\frac{1}{56} \times 6.02 \times 10^{23} \times 1.6 \times 10^{-19})^2}{3^2}$$

2. Superposition: vector sum of the forces

$\Rightarrow |F|$ remains the same if q_1 's sign is changed from negative to positive.



Lecture 2

1. Electric Field

$$\vec{E} \equiv \frac{\vec{F}}{q}$$

* $\equiv$ "is defined as", q : 单位试探电荷

To a point charged particle: $\vec{E} = k \frac{Q}{r^2} \hat{r}$

Superposition $\vec{E} = \sum_i k \frac{Q_i}{r_i^2} \hat{r}_i$

Field points away from positive charges;
toward negative charges

2. Charge density

(1) Linear ($\lambda = \frac{Q}{L}$) C/m

(2) Surface ($\lambda = \frac{Q}{A}$) C/m²

(3) Volume ($\lambda = \frac{Q}{V}$) C/m³

* $A_{\text{sphere}} = 4\pi R^2$ $A_{\text{cylinder}} = 2\pi RL$

$V_{\text{sphere}} = \frac{4}{3}\pi R^3$ $V_{\text{cylinder}} = \pi R^2 L$

$$\vec{E} = \sum_i k \frac{Q_i}{r_i^2} \hat{r}_i \Rightarrow \vec{E} = \int k \frac{dq}{r^2} \hat{r}$$

Lecture 3

Prelecture

Electric Field

$$\vec{E} = k \frac{q}{r^2} \hat{r}$$

Surface Area of Sphere

$$A = 4\pi r^2$$

Density of Field lines:

$$D = \frac{\text{Number of Lines}}{A} = \frac{N}{4\pi r^2}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$N \equiv \frac{q}{\epsilon_0}$$

$$\Rightarrow D = \frac{q}{4\pi\epsilon_0 r^2} = E$$

$$E \propto \frac{N}{4\pi r^2}$$
$$N \propto \frac{q}{\epsilon_0}$$

* $k = \frac{1}{4\pi\epsilon_0}$, ϵ_0 : 真空介电常数

Electric Flux: ϕ 电通量
最简单的情形: $\phi = EA$

$\Rightarrow$ 适用前提 ① 场均匀 ② 面与场垂直

$\Rightarrow$ 通量严格定义: $\phi = \int \vec{E} \cdot d\vec{A}$

$$* d\phi = \vec{E} \cdot d\vec{A}$$

对点电荷: $\phi = EA = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cdot 4\pi r^2 = \frac{q}{\epsilon_0} = N$

半球 $\phi_{\text{半球}} = \frac{1}{2} \frac{q}{\epsilon_0}$, $\phi_{\text{平面}} = 0$

$\Rightarrow$ Gauss' Law $\phi = \frac{q_1}{\epsilon_0} + \frac{q_2}{\epsilon_0} + \dots = \frac{q_{\text{enclosed}}}{\epsilon_0}$

通量是标量, 但带正负号. $\phi > 0$: 总体穿出
 $\phi = 0$ 平衡
 $\phi < 0$: 总体穿入